History of the atomic model 1a

1 Read the following text about scientists and cross out the incorrect verb alternatives. The verbs are underlined. The first one has been done for you. Remember that historical recounts should be written in past tense.

People often ~~made up~~ / make up models to explain / to explained things. Scientists perform / performs experiments which show / shows them how something acts, or what something looks / looked like. This information is used to develop / to develops a model. The model of the atom has underwent / has undergone many changes.

The idea of atoms is suggested / was suggested by Democritus in ancient Greece (460 – 370 BC). He logically deducted / deduced that matter could not continue to be cut into smaller pieces forever, and that eventually there had to be / have to be a smallest particle, which he calls / called the atom (about 400 BC).

In England, John Dalton (1766 – 1844) heard / heared about the experiments of other chemists. From their results he concluded / concludes that the atoms of the same element was / were identical and cannot be / could not be broken. In a chemical reaction, the atoms were separated, joined or rearranged. Dalton's model of the atom were / was like a ball (1803).

When English scientist Joseph Thomson (1856 – 1940), and other scientists, experiment / experimented with electricity, they finded / found that cathode rays are / were made up of tiny particles which travelled / travel in straight lines and carried / carryed a negative electric charge. These cathode rays came / comed from the negative terminal (ie the cathode) and travelled to the positive terminal (ie the anode), when very high voltages have / were applied. Thomson concludes / concluded that the rays make up / were made up of electrons, which were very small (they was/ were 2000 times lighter than a hydrogen atom) and had / have a negative charge. But if negative electrons came / come from neutral and relatively large atoms, the atom had to / has to look like a plum pudding: the dough of the pudding was a uniform substance with a positive charge and contained / contain tiny sultanas with a negative charge (1904).

2 In the following text about atoms and radioactivity, there are 12 spelling mistakes. Underline the incorrectly spelt words and write them correctly in the margin.

Radioactivity had been discovered and alpha particles were identified as tiny positive protons. A student of Joseph Thomson, the New Zealand born Ernest Rutherford (1871 – 1937), shot a beam of alpha particals at a very thin sheet of gold foil. He expected the protons to fly straight threw the foil, practically uneffected by the positive and negative particles in the gold atoms. Rutherford was suprised to find that some protons were deflected, some even bounced back to the sauce. Based on his measurements, Rutherford concluded that the atom's protons had to be concentrated in a small nucleus leaving the electrons to suround it. The mass of the atom was determined by the nucleus, which was much heavier than the electrons, and its size was determined by the electrons, which moved around the nucleus. Rutherford's model of an atom looked like a spherical cloud of negative electrons with a positive nucleus in the centre. If the atom was as big as a football feild, the nucleus would be as big as a tennis ball (1911).

The German Max Planck (1858 – 1947) formulated the quantum theory: Energy is given out or taken in by atoms only in lots (called quanta). Niels Bohr from Denmark (1885 – 1962) concluded that electrons jumped from one energy level to another, each jump being worth one qantum of energy. As a result the atomic model became like a soler system: a small nucleus, containing the heavy positive protons, was surrounded by several orbits on which the tiny negative electrons circled the nucleus (1913).

Erwin Schrodinger (1887 – 1961) and Wolfgang Pauli (1900 – 1958), both from Austria, proposed another model for the atom: The electron shells contained subshells of various shapes, each with a maximum of two electrons. But it was impossible to state where a particular electron was at any time; one could only say where an electron was most likley to be. Thus the atom appered like a nucleus inside many electron clouds (1926).

When the English scientist James Chadwick (1891 – 1974) discovered the existance of neutrons, which were as heavy as protons but carried no charge, the atomic model changed to include neutrons inside the nucleus together with the protons (1932).

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History of the atomic model 1b

3 The following words are snippets from sentences which summarise the text from questions 1 and 2. Sort the snippets from each sentence and then write down the sensible sentences.

a | atom | Democritus from ancient Greece | named | in about 400BC | the smallest particle |.

b | believed | in 1803 | like a ball | that | the atom | The English scientist John Dalton | was |.

c | drew | In 1904 | Joseph Thomson from England | like a positively charged plum pudding with negative plums in it | the atom |.

d | and | concentrated | Ernest Rutherford | In 1911 | in the central nucleus | moved | the negative electrons | onto a cloud surrounding the nucleus | the positive charge |, who was born in New Zealand, |.

e | but | in 1913 | kept | onto several orbits | placed | rather like our solar system | The Danish Niels Bohr | the negative electrons | the positive nucleus |.

f | an atom where the nucleus was surrounded by several orbital clouds | and | combined | developed | in 1926 | The Austrian scientists Erwin Schrodinger and Wolfgang Pauli | the idea of orbits and clouds |.

g | discovered | In 1932 | inside the positive nucleus | neutral neutrons | the English scientist James Chadwick |.

Drawing atomic models

In the paragraphs above various scientists developed certain models of atoms. Each model explained what the scientists knew at that time. New discoveries made new atomic models necessary.

4 Draw each model as described in the text. Next to your diagram write the description of the model as well as the name of the scientist and the year of discovery.

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